Probability, Statistics and Random Processes

Third Edition

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Tata McGraw-Hill Publishing Company Limited
NEW DELHI

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San Juan Santiago Singapore Sydney Tokyo Toronto

Tata McGraw-Hill

Published by the Tata McGraw-Hill Publishing Company Limited,
7 West Patel Nagar, New Delhi 110 008.

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This edition can be exported from India only by the publishers,
Tata McGraw-Hill Publishing Company Limited.

ISBN 13: 978-0-07-066925-3
ISBN 10: 0-07-066925-2

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Published by the Tata McGraw-Hill Publishing Company Limited, 7 West Patel Nagar, New Delhi 110 008, typeset at Bukprint India, B-180A, Guru Nanak Pura, Laxmi Nagar-110 092 and printed at Adarsh Printers, C-50-51, Mohan Park, Naveen Shahdara, Delhi 110 032

Cover: SDR Printers

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HYPOTHESIS TESTING – II

Small Sample Tests

In statistical analysis, **small sample tests** are used when the size of the sample is small.

- A **large sample** is one where

$$n \geq 30$$

- A **small sample** is one where

$$n < 30$$

The most important tests used for small samples are:

- **Student's t-test**
- **F-test**

Student's t-Test

When the **population standard deviation is not known** and the **sample size is less than 30**, the **t-test** is used.

The t-distribution is also called the **Student's t-distribution**.

Let

$$x_1, x_2, \ldots, x_n$$

be a random sample from a normal population with mean μ and variance σ^2.

The test statistic is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

t-Distribution

A random variable X follows a **t-distribution** if its probability density function is

$$f(t) = \frac{k}{\left(1 + \frac{t^2}{v}\right)^{(v+1)/2}}, \qquad -\infty < t < \infty$$

where

- v = degrees of freedom
- k = constant

The constant k is chosen such that

$$\int_{-\infty}^{\infty} f(t)\, dt = 1$$

After simplification

$$k = \frac{1}{\sqrt{v\pi}} \frac{\Gamma\left(\frac{v+1}{2}\right)}{\Gamma\left(\frac{v}{2}\right)}$$

Degrees of Freedom

The **degrees of freedom** represent the number of independent values that can vary in a statistical calculation.

For a sample of size n,

$$v = n - 1$$

Assumptions of t-Distribution

1. The population from which the sample is drawn is **normal**.
2. The sample is **random**.
3. The **sample size is less than 30**.
4. The **population standard deviation is unknown**.

Properties of t-Distribution

1. The probability curve is **symmetrical**.
2. The tails approach the x-axis asymptotically.
3. When $n \to \infty$, the t-distribution approaches the **normal distribution**.
4. The shape of the distribution depends on the **degrees of freedom**.

Applications of t-Distribution

The t-test is used to test the significance of the difference between

1. The mean of a **small sample and the population mean**.
2. The means of **two small samples**.

Test of Significance for Single Sample Mean

To test whether the sample mean differs from the population mean:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

Degrees of freedom

$$v = n - 1$$

Compare calculated t with tabulated value t_α.

Decision rule:

- If $|t| < t_\alpha$ → accept H_0
- If $|t| > t_\alpha$ → reject H_0

Confidence Interval for Population Mean

The confidence interval for the mean is

$$\bar{x} \pm t_\alpha \frac{s}{\sqrt{n}}$$

Example 1

A machinist produces engine parts with axle diameter **0.7 inches**.

Sample data

$$n = 10, \quad \bar{x} = 0.742, \quad s = 0.04$$

Test at **5% level**.

Hypotheses

$$H_0 : \mu = 0.7$$

$$H_1 : \mu \neq 0.7$$

Test statistic

$$t = \frac{0.742 - 0.7}{0.04/\sqrt{10}} = 3.15$$

Degrees of freedom

$$v = 9$$

Critical value

$$t_{0.025,9} = 2.26$$

Since

$$|t| > 2.26$$

Reject H_0.

Conclusion: There is significant difference from specification.

Example 2

Expected diameter = **1.75 cm**

Sample:

$$\bar{x} = 1.85, \quad s = 0.1, \quad n = 10$$

$$t = \frac{1.85 - 1.75}{0.1/\sqrt{10}} = 3$$

Critical value

$$t_{0.025,9} = 2.26$$

Reject H_0.

Conclusion: Machinist work is **not meeting specification**.

Example 3

Oil tins sample

$$n = 10, \quad \bar{x} = 15.8, \quad s = 0.5$$

Test

$$H_0 : \mu = 16$$

$$t = \frac{15.8 - 16}{0.5/\sqrt{10}} = -1.2$$

$$|t| < 2.26$$

Accept H_0.

Conclusion: Mean weight does not differ significantly from 16 kg.

Example 5 — Packing Machine Test

Sample weights

0.49, 0.48, 0.47, 0.48, 0.49, 0.50, 0.51, 0.49, 0.48, 0.50, 0.51, 0.48

Sample mean

$$\bar{x} = 0.49$$

Variance

$$s^2 = 0.00016364$$

Standard deviation

$$s = 0.0128$$

Hypothesis

$$H_0 : \mu = 0.5$$

Test statistic

$$t = \frac{0.49 - 0.5}{0.0128/\sqrt{12}} = -2.59$$

Critical value

$$t_{0.025,11} = 2.20$$

Since

$$|t| > 2.20$$

Reject H_0.

Conclusion: Packing mean is **significantly different from 0.5 kg**.

Example 6 — Blood Sugar Test

Sample data

70, 120, 110, 101, 88, 83, 95, 107, 100, 98

Sample mean

$$\bar{x} = 97.2$$

Hypothesis

$$H_0 : \mu = 100$$

Test statistic

$$t = -0.6203$$

Critical value

$$t_{0.025,9} = 2.262$$

Accept H_0.

Confidence interval

$$87.107 < \mu < 107.40$$

Difference Between Means of Two Small Samples

Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where pooled variance

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

Degrees of freedom

$$v = n_1 + n_2 - 2$$

Example — Two Sample Means

Sample	Size	Mean	SD
1	16	23.4	2.5
2	12	24.9	2.8

Pooled variance

$$s^2 = 7.46$$

$$s = 2.732$$

Test statistic

$$t = -1.4378$$

Critical value

$$t_{0.025,26} = 2.06$$

Accept H_0.

Conclusion: Means are not significantly different.

Diet Experiment Example

Diet A mean

$$6.4$$

Diet B mean

$$5$$

Pooled variance

$$s^2 = 12.968$$

$$t = 0.82$$

Critical value

$$t_{0.05,16} = 1.75$$

Accept H_0.

Conclusion: Diet A is **not significantly better** than Diet B.

Battery Life Example

Battery A

$$n_A = 14, \quad \bar{x}_A = 94, \quad s_A = 16$$

Battery B

$$n_B = 13, \quad \bar{x}_B = 86, \quad s_B = 20$$

Test statistic

$$t = 1.0986$$

Critical value

$$t_{0.025,25} = 2.060$$

Accept H_0.

Conclusion: No significant difference between battery lives.

Paired t-Test

When observations occur in **pairs**, use paired t-test.

Test statistic

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

where

$$\bar{d} = \frac{\sum d_i}{n}$$

$$s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1}$$

Example — Meditation Experiment

Before

10, 15, 9, 3, 7, 12, 16, 17, 4

After

12, 17, 8, 5, 6, 11, 18, 20, 3

Mean difference

$$\bar{d} = -0.778$$

Variance

$$s^2 = 2.945$$

Standard deviation

$$s = 1.716$$

Test statistic

$$t = -1.36$$

Critical value

$$t_{0.025,8} = 2.31$$

Accept H_0.

Conclusion: Meditation course **is not statistically effective**.